

Floating-point number parsing with perfect accuracy at a gigabyte per second

Daniel Lemire

professor, Université du Québec (TÉLUQ)

Montreal 🇨🇦

blog: <https://lemire.me>

X: [@lemire](#)

GitHub: <https://github.com/lemire/>

! work with Michael Eisel, Ivan Smirnov, Nigel Tao, R. Oudompheng, Carl Verret and others!

How fast is your disk?

- PCIe 4 disks: 5 GB/s reading speed (sequential)
- PCIe 5 disks: 14.5 GB/s (0.20\$/GB)

Fact

Single-core processes are often CPU bound

How fast can you ingest data?

```
{ "type": "FeatureCollection",  
  "features": [  
    [[-65.6136169999999977, 43.42027300000000009],  
     [-65.6197200000000029, 43.4180529999999986],  
     [-65.625, 43.42137900000000059],  
     [-65.63612399999999882, 43.4497149999999969],  
     [-65.6330569999999951, 43.47470900000000132],  
     [-65.61138900000000031, 43.51305400000000068],  
     [-65.60583500000000013, 43.5161059999999979],  
     [-65.598343, 43.5158309999999935],  
     [-65.56610100000000003, 43.50833100000000055],  
     ...  
  ]  
}
```

How fast can you parse numbers?

```
std::stringstream in(mystring);  
while(in >> x) {  
    sum += x;  
}  
return sum;
```

50 MB/s (Linux, GCC -O3)

Source: <https://lemire.me/blog/2019/10/26/how-expensive-is-it-to-parse-numbers-from-a-string-in-c/>

Some arithmetic

5 GB/s divided by 50 MB/s is 100.

Got 100 CPU cores?

Want to cause climate change all on your own?

How to go faster?

- Fewer instructions (simpler code)
- Fewer branches

How fast can you go?

AMD Rome (Zen 2). GNU GCC 10, -O3.

function	bandwidth	instructions	ins/cycle
strtod (GCC 10)	200 MB/s	1100	3
ours	1.1 GB/s	280	4.2

17-digit mantissa, random in [0,1].

Floats are easy

- Standard in Java, Go, Python, Swift, JavaScript...
- IEEE standard well supported on all recent systems
- 64-bit floats can represent all integers up to 2^{53} exactly.

Floats are hard

```
> 0.1 + 0.2 == 0.3  
false
```

Generic rules regarding "exact" IEEE support

- Always round to nearest floating-point number (*,+,/)
- Resolve ties by rounding to nearest with an even decimal mantissa/significand.

Benefits

- Predictable outcomes.
- Debuggability.
- Cross-language compatibility (same results).

Challenges

- Machine A writes float x to string
- Machine B reads string gets float x'
- Machine C reads string gets float x''

Do you have $x = x'$ and $x = x''$?

What is the problem?

Need to go from

$$w \times 10^q$$

(e.g., 123e5)

to

$$m \times 2^p$$

Example

$$0.1 \rightarrow 7205759403792793 \times 2^{-56}$$

0.10000000000000000000555

$$0.2 \rightarrow 7205759403792794 \times 2^{-55}$$

0.2000000000000000000111

$$0.3 \rightarrow 5404319552844595 \times 2^{-54}$$

0.29999999999999999998889776975

Problems

Start with 32323232132321321111e124.

Lookup 10^{124} as a float (not exact)

Convert 32323232132321321111 to a float (not exact)

Compute $(10^{124}) \times (32323232132321321111)$

Approximation $\times$ Approximation = Even worse approximation!

Insight

You can always represent floats exactly (binary64) using at most 17 digits.

Never to this:

3.14159265358979323846264338327950288419716939937510582097494459230781
64062862089986280348253421170679

WHAT THE NUMBER OF DIGITS IN YOUR COORDINATES MEANS

LAT/LON PRECISION	MEANING
28°N, 80°W	YOU'RE PROBABLY DOING SOMETHING SPACE-RELATED
28.5°N, 80.6°W	YOU'RE POINTING OUT A SPECIFIC CITY
28.52°N, 80.68°W	YOU'RE POINTING OUT A NEIGHBORHOOD
28.523°N, 80.683°W	YOU'RE POINTING OUT A SPECIFIC SUBURBAN CUL-DE-SAC
28.5234°N, 80.6830°W	YOU'RE POINTING TO A PARTICULAR CORNER OF A HOUSE
28.52345°N, 80.68309°W	YOU'RE POINTING TO A SPECIFIC PERSON IN A ROOM, BUT SINCE YOU DIDN'T INCLUDE DATUM INFORMATION, WE CAN'T TELL WHO
28.5234571°N, 80.6830941°W	YOU'RE POINTING TO WALDO ON A PAGE
28.523457182°N, 80.683094159°W	"HEY, CHECK OUT THIS SPECIFIC SAND GRAIN!"
28.523457182818284°N, 80.683094159265358°W	EITHER YOU'RE HANDING OUT RAW FLOATING POINT VARIABLES, OR YOU'VE BUILT A DATABASE TO TRACK INDIVIDUAL ATOMS. IN EITHER CASE, PLEASE STOP.

credit: xkcd

We have 64-bit processors

So we can express all positive floats as

12345678901234567E+/-123 .

Or $w \times 10^q$

where mantissa $w < 10^{17}$

But 10^{17} fits in a 64-bit word!

Factorization

$$10 = 5 \times 2$$

Overall algorithm

- Parse decimal mantissa to a 64-bit word!
- Precompute 5^q for all powers with up to 128-bit accuracy.
- Multiply!
- Figure out right power of two

Tricks:

- Deal with "subnormals"
- Handle excessively large numbers (infinity)
- Round-to-nearest, tie to even

SIMD

- Stands for Single instruction, multiple data
- Allows us to process 16 bytes or more with one instruction
- Supported on all modern CPUs (phone, laptop)
- Not portable

SWAR

- Stands for SIMD within a register
- Use normal instructions, portable (in C, C++,...)
- A 64-bit register can be viewed as 8 bytes
- Requires some cleverness

Check whether we have a digit

In ASCII/UTF-8, the digits 0, 1, ..., 9 have values 0x30, 0x31, ..., 0x39.

To recognize a digit:

- The high nibble should be 3.
- The high nibble should remain 3 if we add 6 (0x39 + 0x6 is 0x3f)

Silly formula to recognize a digit

- $(x \& 0xF0) + (((x + 6) \& 0xF0) \gg 4) = 0x33$

Check whether we have 8 consecutive digits

```
bool is_made_of_eight_digits_fast(const char *chars) {  
    uint64_t val;  
    memcpy(&val, chars, 8);  
    return (((val & 0xF0F0F0F0F0F0F0F0) |  
            (((val + 0x0606060606060606) & 0xF0F0F0F0F0F0F0F0) >> 4))  
            == 0x3333333333333333);  
}
```

(Works with ASCII, harder if input is UTF-16 as in Java/C#)

Then construct the corresponding integer

Using only three multiplications (instead of 7):

```
uint32_t parse_eight_digits_unrolled(const char *chars) {  
    uint64_t val;  
    memcpy(&val, chars, sizeof(uint64_t));  
    val = (val & 0x0F0F0F0F0F0F0F0F) * 2561 >> 8;  
    val = (val & 0x00FF00FF00FF00FF) * 6553601 >> 16;  
    return (val & 0x0000FFFF0000FFFF) * 42949672960001 >> 32;  
}
```

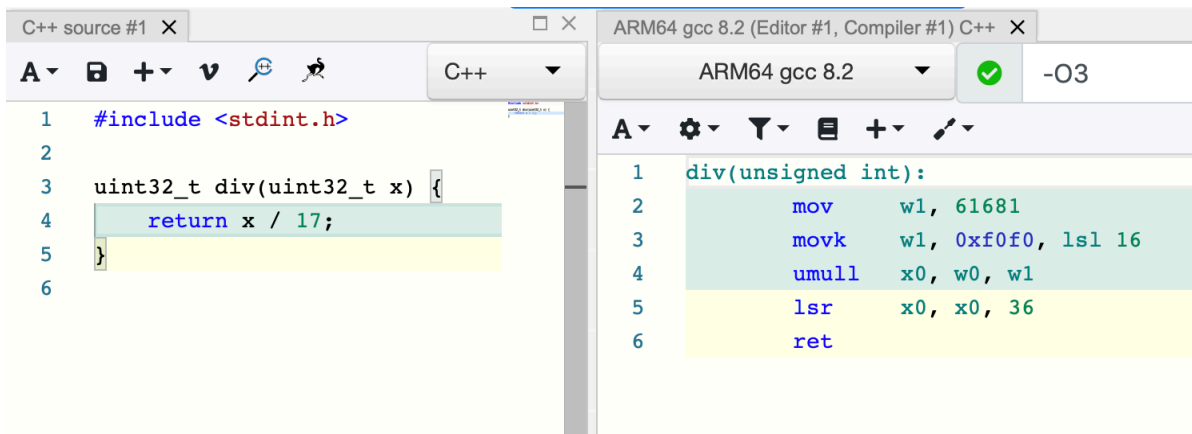
Positive powers

- Compute $w \times 5^q$ where 5^q is only approximate (128 bits)
- Maybe surprisingly, 128-bit precision is all that is needed to always get exact results.

Noble Mushtak, Daniel Lemire, Fast Number Parsing Without Fallback Software: Practice and Experience 53 (7), 2023

Negative powers

- Compilers replace division by constants with multiply and shift



```
C++ source #1 X
1 #include <stdint.h>
2
3 uint32_t div(uint32_t x) {
4     return x / 17;
5 }
6

ARM64 gcc 8.2 (Editor #1, Compiler #1) C++ X
ARM64 gcc 8.2 -O3
1 div(unsigned int):
2     mov     w1, #61681
3     movk    w1, #0xf0f0, lsl #16
4     umull   x0, w0, w1
5     lsr     x0, x0, #36
6     ret
```

credit: godbolt

Reading: Integer Division by Constants: Optimal Bounds,
<https://arxiv.org/abs/2012.12369>

Negative powers

- Precompute $2^b/5^q$ (reciprocal, 128-bit precision)
- Always get exact results.

What about tie to even?

- Need absolutely exact mantissa computation, to infinite precision.
- But only happens for small decimal powers ($q \in [-4, 23]$) where absolutely exact results are practical.

What if you have more than 19 digits?

- Truncate the mantissa to 19 digits, map to w .
- Do the work for $w \times 10^q$
- Do the work for $(w + 1) \times 10^q$
- When get same results, you are done. (99% of the time)

Overall

- With 64-bit mantissa.
- With 128-bit powers of five.
- Can do exact computation 99.99% of the time.
- Fast, cheap, accurate.

Full product?

- 64-bit $\times$ 64-bit $\rightarrow$ 128-bit product
- GNU GCC: `__uint128_t` .
- Microsoft Visual Studio: `_umul128`
- ARM intrinsic: `__umulh`
- Go: `bits.Mul64`
- C#: `Math.BigMul`

Leading zeros

- How many consecutive leading zeros in 64-bit word?
- GNU GCC: `__builtin_clzll`
- Microsoft Visual Studio: `_BitScanReverse64`
- C++20: `std::countl_zero`
- Go: `bits.LeadingZeros64`
- C#: `BitOperations.LeadingZeroCount`

C/C++

- https://github.com/lemire/fast_float
- GNU GCC
- LLVM clang
- used by Apache Arrow, Yandex ClickHouse, Microsoft LightGBM



Go

- Algorithm adapted to Go's standard library (ParseFloat) by Nigel Tao and others
- Release notes (version 1.16): *ParseFloat (...) improving performance by up to a factor of 2.*
- Perfect rounding.
- Blog post by Tao: [The Eisel-Lemire ParseNumberF64 Algorithm](#)

Rust

function	speed
from_str (standard)	130 MB/s
lexical (popular lib.)	370 MB/s
fast-float	1200 MB/s

R

rcppfastfloat: <https://github.com/eddelbuettel/rcppfastfloat>

3x faster than standard library

C#

`FastFloat.ParseDouble` is 5x faster than standard library (`Double.Parse`)

<https://github.com/CarlVerret/csFastFloat/>

credit: Carl Verret, Egor Bogatov (Microsoft) and others

Further reading

- Noble Mushtak, Daniel Lemire, Fast Number Parsing Without Fallback, Software: Practice and Experience 53 (7), 2023
- Daniel Lemire, Number Parsing at a Gigabyte per Second, Software: Practice and Experience 51 (8), 2021
- Blog: <https://lemire.me/blog/>